

Konstantinos Roumpas

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Profile

PhD Candidate in Computer Engineering and Informatics researching Human-Computer Interaction, Cognitive Modeling, and AI. I build adaptive systems that combine machine learning, eye-tracking, and multimodal data, and I currently lead technical delivery on an EU Horizon research team. Comfortable moving between hands-on Python/ML work and coordinating a team toward shared research goals.

Education

University of Patras

PhD Candidate in Computer Engineering and Informatics 2023–Present

Research focus: Human-Computer Interaction, Machine Learning, Cognitive Modeling, Adaptive Systems

Aristotle University of Thessaloniki

BSc in Physics 2016–2022

Specialization: Computational Physics, Data Analysis, Non-linear Systems

Research Experience

Frontier Innovations – Researcher / Technical Manager 2025–Present

- Lead technical direction and coordinate a team of researchers and engineers across multiple EU Horizon-funded projects.
- Set research strategy and oversee delivery, translating project objectives into working AI/ML systems.
- Design and develop research-driven GUIs and dashboards that integrate outputs from each project's modules into a unified interface.

HCI & Software Quality Lab, University of Patras – PhD Researcher 2024–2025

- Developed machine learning and large language models (LLMs) for adaptive user interfaces using multimodal signals (eye-tracking, behavioral data, interaction logs).
- Researched AI-assisted systems for education and cognitive state recognition.
- Designed and evaluated prototypes leveraging predictive modeling and deep learning for user behavior analysis.
- Published research in peer-reviewed journals and international AI conferences.

ATERMON B.V. – AI-driven Learning Experience Designer 2024

- Designed and implemented AI-powered learning modules integrating data-driven personalization.
- Collaborated on EU-funded R&D projects in digital learning, inclusion, and human-AI collaboration.
- Applied NLP tools and user analytics to optimize content adaptation and feedback.

Asserted Knowledge O.E. – Instructional Designer & Developer 2021–2024

- Built and evaluated interactive systems using AI-driven behavior modeling and engagement metrics.
- Conducted data analysis with Python, Pandas, and TensorFlow for learning analytics research.
- Authored reports and technical documentation for EU innovation projects involving AI and data science.

Technical Skills

Programming: Python (NumPy, Pandas, TensorFlow, PyTorch), JavaScript, Unity, React

AI/ML: Deep Learning, NLP, Predictive Modeling, Multimodal Data Processing, Eye-tracking Analytics, LLMs

Tools: Jupyter, Git, Docker, Figma, MATLAB

Languages: Greek (native), English (fluent)

Selected Publications

Journal Articles

- [J3] Roumpas, K. et al. (2025). *A Framework for Ethical, Cognitive-Aware Human-Ai Interaction in Multimodal Adaptive Learning Systems* SSRN.
- [J2] Roumpas, K. et al. (2025). *A comparative study of physics-informed and conventional neural networks for predicting on-screen gaze points from eye-tracking data*. Expert Systems with Applications, 281, 127655.
- [J1] Minas, D., Theodosiou, E., Roumpas, K., Xenos, M. (2025). *Adaptive Real-Time Translation Assistance Through Eye-Tracking*. AI, 6(1), 5.

Conference Papers

- [C2] Roumpas, K., Xenos, M. (2025). *Bridging the Gap Between Educators and Game-Based Learning: An AI-Assisted Course Creation Tool*. 3rd Cognitive Models and Artificial Intelligence Conference (AICCONF), 1–7.
- [C1] Roumpas, K., Xenos, M. (2024). *Design of an AI-Assisted Platform to Facilitate Educators in Creating Adaptive Courses*. WAIE 2024, 197–201.